

No. of Functional Features Included in the Solution

Date	28 June 2026
Team ID	XXXXXX
Project Name	Smart Agricultural Production Optimization Engine (OptiCrop)
Maximum Marks	5 Marks

Functional Features Overview:

This document lists all the functional features included in the OptiCrop project solution. Each feature is described with its purpose, implementation status, and contribution to the overall solution.

S. No	Feature Name	Feature Description	Module / Component	Status	Marks Contribution
1	Crop Recommendation Engine	Accepts N, P, K, temperature, humidity, pH, and rainfall inputs and predicts the most suitable crop using a trained ML model	ML Model / Backend	Done	1.0
2	Soil & Climate Input Form	Web-based input form for users to enter all 7 environmental parameters with validation and range hints	Frontend / UI	Done	0.5
3	Input Validation & Error Handling	Server-side validation of all input fields ensuring values are numeric and within acceptable agronomic ranges; returns descriptive error messages	Backend / Flask	Done	0.5
4	Exploratory Data Analysis (EDA)	Visual analysis of the crop dataset using Matplotlib and Seaborn — includes heatmaps, box plots, pairplots, and class distribution charts	Data Analysis / Jupyter	Done	0.5
5	ML Model Training & Evaluation	Trains a Random Forest classifier on the crop dataset; evaluates using accuracy, precision, recall, F1-score, and confusion matrix	ML Engine / Scikit-learn	Done	1.0
6	REST API for Prediction	Flask /predict POST endpoint that receives JSON input, runs ML inference, and returns crop recommendation with confidence score as JSON response	Backend / Flask API	Done	0.5
7	Dynamic	JavaScript renders the crop	Frontend / JS	Done	0.5

	Result Display	recommendation result dynamically on the page after prediction, with smooth scroll and loading spinner			
8	Data Preprocessing Pipeline	Handles missing values, encodes crop labels using LabelEncoder, scales features using StandardScaler, and splits data into train/test sets	Data Processing / Pandas	Done	0.5

Feature Summary:

Metric	Count / Value
Total Features Planned	8
Total Features Implemented	7
Core / Must-Have Features	5 (Crop Recommendation, Input Form, Validation, ML Model, REST API)
Additional / Nice-to-Have Features	3 (EDA, Dynamic Display, Preprocessing Pipeline)
Features Tested & Verified	0

Feature Category Breakdown:

S. No	Category	Features in Category	Example Features
1	User Interface (UI)	2	Soil & Climate Input Form, Dynamic Result Display
2	Backend / Logic	2	REST API for Prediction, Input Validation & Error Handling
3	Database / Storage	1	Data Preprocessing Pipeline (CSV dataset loading and management)
4	API / Integration	1	Flask REST API /predict endpoint
5	Machine Learning / Analytics	2	ML Model Training & Evaluation, Exploratory Data Analysis